

Mem0 Integration & 3-Tier Mobile MoE Architecture

1. The Mobile RAM Ceiling (<5.0 GB)

Running heavy vector search alongside foundation models exhausts mobile memory. The 3-Tier MoE architecture splits reasoning from persistence:

- **Tier 1 (Triage Router - Qwen 3.5 0.8B)**: Loads in ~1.2 GB RAM. Parses raw text input, determines intent, and calls `Mem0.search()` for entity facts.
- **Tier 2 (Core Executor - Qwen 3.5 4B / 9B)**: Loads in ~3.8 GB RAM. Receives concise, pre-filtered context without full chat history bloat, generating code/tools.
- **Tier 3 (Persistence Layer - Mem0 + SQLite)**: Updates episodic memory logs asynchronously using `Mem0.add()`.